

Jargon Buster

Encryption: Performing mathematical operations on messages to hide their content.

Decryption: The reverse process of encryption—getting the original message back from the encrypted version.

Plaintext: A clear, un-encrypted message, like “Hello.”

Cipher: An algorithm used to encrypt

plaintext and convert it into seemingly nonsensical *ciphertext*.

Key: A string of numbers that is given to the cipher along with the message. The cipher’s resulting output depends on the key.

Ciphertext: The final result of plaintext fed to the cipher along with the key, for example, æW æÇ^[Å.

Light behaves strangely in the sub-atomic world. Sometimes it’s a wave, and sometimes it’s a stream of particles called photons. It is these photons that will be used to carry our data in the future.

In the world of classical physics, a bit is a bit. It’s *tangible*—either as an electrical signal in a wire or the intensity of a light beam in a fibre-optic cable. And because it’s tangible, it can be measured. Measuring something like the electrical current through a wire doesn’t affect the current itself—it just passes through unchanged. This means that someone could tap a communication line without either party ever finding out.

In the quantum world, though, things are a little different. There are properties of photons that are tied together so tightly that measuring one property will cause a change in the others, disturbing the entire system. This way, any attempt to eavesdrop would easily be detected.

The Crazy, Crazy Quantum World

Suppose you have a friend behind you. You can’t see him, so he could be standing, sitting, running, jumping, anything. Of course, you could just turn around and take a look, but do you really know what he was doing the instant before you looked? If you lived in the sub-atomic world, he was actually doing *everything* till you observed him; once seen, he settled into the state you saw him in. What state? Well, you could see him in any state at all—it’s random. Why random? Ah, well, Einstein himself couldn’t figure it out in his lifetime, so don’t expect an explanation from us.

Even though quantum randomness gives scientists sleepless nights, it actually helps cryptographers sleep better. The problem with the one-time pad is that in computers, nothing is really random—programs are written that *simulate* randomness, but are actually predictable.

Er...

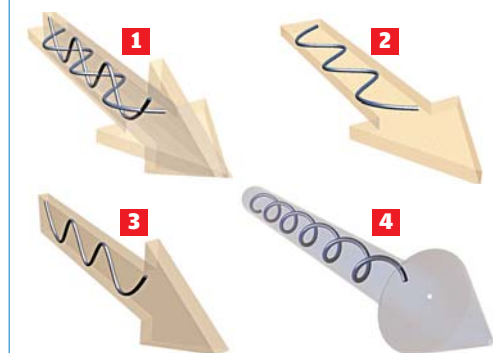
At the core of quantum physics is the concept of *Uncertainty*. Two properties of a photon can be related in such a way that the more certain you get about one, the less certain you get about another. *Heisenberg’s Uncertainty Principle* describes the relationship between the position and momentum of particles. Basically, if you know the position of a photon, you will never know its momentum unless you are prepared to give up your knowledge of its position.

Stand Me Up, Lie Me Down, Turn Me Around

The property of photons that is used in quantum cryptography is *Polarisation*. Light waves move in different *planes*—vertical, horizontal, and everything in between. *Linear Polarisation* is the act of selecting only those light waves that move in one particular plane. Light could either be vertically or horizontally polarised. There is also the special case of polarisation called *Circular Polarisation*, where the light wave is made to move in a circular fashion rather than the classical up-and-down in one plane. Circular polarisation could happen in the right or left direction. Each photon carries with it information about its polarisation. Linear and circular polarisations are bound by uncertainty—your measurement of linear polarisation will destroy your measurement of circular polarisation, and vice versa.

Much to our relief, quantum cryptography can be understood even if you don’t understand polarisation.

Polarising Light



1. Light travels in all planes. 2. Horizontal Polarisation—isolating light that travels horizontally
3. Vertical Polarisation 4. Circular polarisation

Setting The Scene—Artur Ekert’s Card Tricks

In 1991, Artur Ekert came up with an interesting analogy to help us lesser mortals understand quantum cryptography.

Suppose we have a whole bunch of sealed envelopes with either blue or red cards in them. Each card has either 0 or 1 written on it. Blue cards can only be read using a special “blue machine,” and red cards can only be read using a “red machine.” If we put a blue card in a red machine, the machine will throw out a random guess as to whether there was a 0 or 1 written on the card, and vice versa.

The colour of the card (the photon) represents its polarisation—let’s say red for linear and blue for circular. A 0 on a red card would mean that the photon is horizontally polarised, and a 1 would mean it is vertically polarised. Similarly, a 0 on a blue card would mean left-circular polarisation and a 1 would mean right-circular polarisation.

Somewhere in these observations comes the whole craziness of the quantum world. If we put an envelope with a blue card that says “1” into